

## Data Flow Diagram

|               |                                                   |
|---------------|---------------------------------------------------|
| Date          | 03-07-2026                                        |
| TeamID        |                                                   |
| Project Name  | Smart Agricultural Production Optimization Engine |
| Maximum Marks | 2 Marks                                           |

### Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system between external entities, processes, and data stores. Use the symbol legend below as a guide.

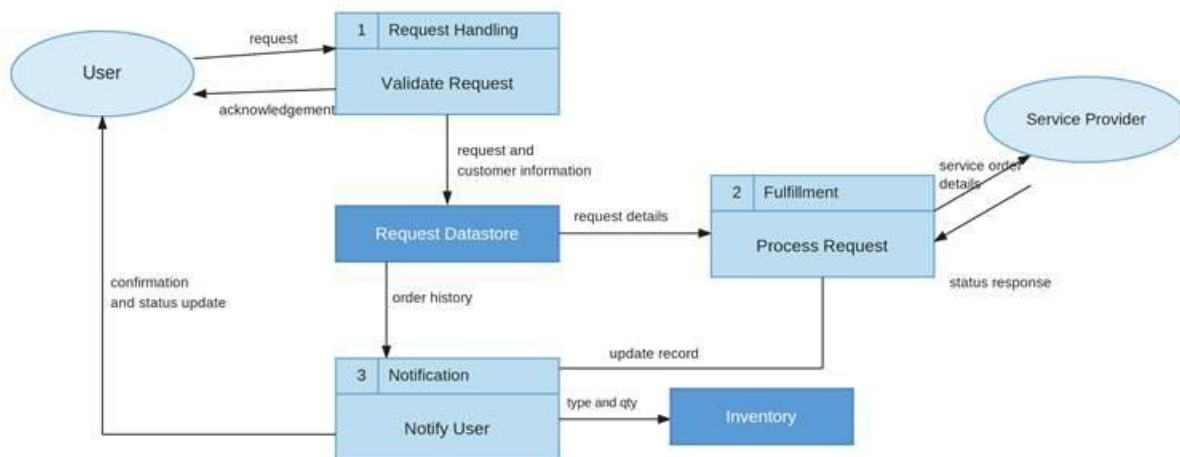


DFD Symbol Legend

| Symbol                            | Name            | Description                                                                                                          |
|-----------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------|
| Oval / Rounded shape              | External Entity | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| Rectangle with numbered header    | Process         | An activity that transforms incoming data into outgoing data (e.g., Value Date Request, Process Order).              |
| Rectangle (solid fill, no number) | Data Store      | A place where data is held for later use (e.g., Customer Database, Inventory).                                       |
| Labeled arrow                     | Data Flow       | The movement of data between entities, processes, and data stores, labeled with what data is being passed.           |

### Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



## Level 0 Data Flow Diagram

Project: Smart Agricultural Production Optimization Engine

### Main Process:

1.0 Opt i Crop Engine - Farmer's suitable crop suggestion system

### External Entities:

1. Farmer - Location and soil details input chest
2. Weather API - Rainfall and temperature data provider
3. Govt Agriculture DB - Soil type and market price data provider

### Data Stores:

1. D1: Farmer Profile Database - Farmer details store
2. D2: ML Model Database - AI prediction model store

### Data Flows:

1. Farmer → 1.0 Opt i Crop Engine : Location + Soil Type
2. Weather API → 1.0 Opt i Crop Engine : Weather Data
3. Govt Agri DB → 1.0 Opt i Crop Engine : Soil & Price Data
4. 1.0 Opt i Crop Engine → Farmer : Crop Suggestion + Fertilizer Plan
5. 1.0 Opt i Crop Engine → D1 : Save Farmer Data
6. D2 → 1.0 Opt i Crop Engine : Prediction Result

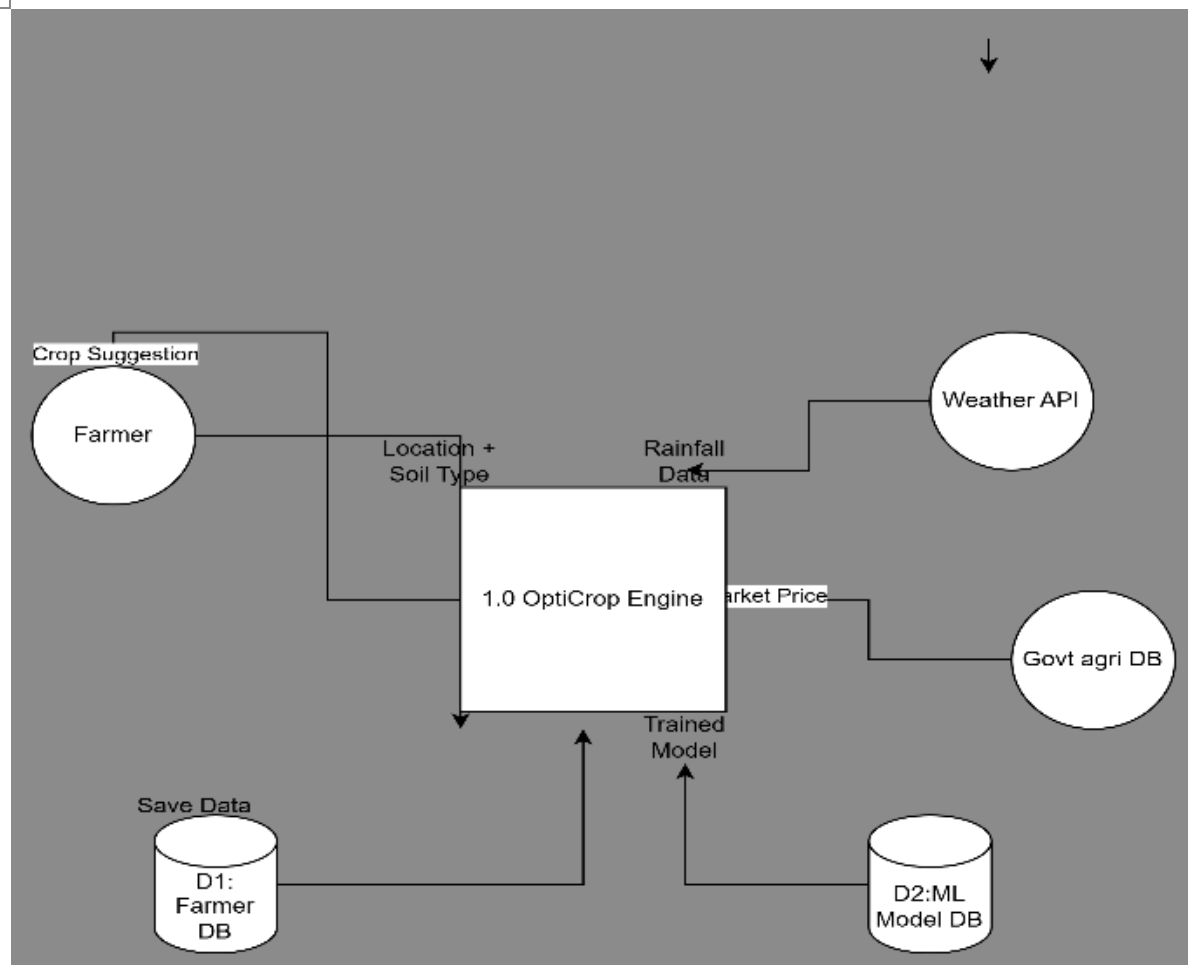


Figure1 : Level 0 DFD for Smart Agricultural Production Optimization Engine